AI MINI-CAPSTONE PROJECT REPORT

Prompt Engineering Strategies Compared: Zero-Shot, Few-Shot and Chain-of-Thought Prompting

Course Code: Artificial Intelligence (25CS301) • Department of Computer Science & Engineering

Academic Year: 2026–2027	**Target LLM:** Google Gemini API (gemini-2.5-flash)
Framework: Streamlit, FastAPI, Plotly	**Dataset:** 25 Curated Questions across 5 Domains
Evaluation: Multi-Metric Ground Truth	**Deployment:** Vercel & Streamlit Cloud

1. Problem Statement

Modern Large Language Models (LLMs) often exhibit unpredictable behaviors, hallucinations, and non-deterministic formatting when prompted naively. While prompt engineering has emerged as the primary mechanism for interacting with generative models, developers face critical trade-offs: Zero-Shot queries are fast and cheap but frequently fail on multi-step reasoning; Few-Shot prompting improves formatting but introduces token overhead; and Chain-of-Thought (CoT) prompting achieves deductive precision at the expense of substantial latency and API token cost. There is a lack of rigorous, empirical comparative benchmarking contrasting these strategies across standardized cognitive categories.

2. Objectives

- **Architect a Comparative Benchmark:** Implement a modular, interactive system evaluating Zero-Shot, Few-Shot, and Chain-of-Thought prompts.
- **Controlled Multi-Strategy Execution:** Simultaneously dispatch identical questions to the Google Gemini API with invariant model parameters.
- **Multi-Dimensional Objective Evaluation:** Systematically quantify answer correctness, topical relevance, reasoning depth, latency, and token efficiency.
- **Standardized Dataset Testing:** Benchmark 25 curated test cases across 5 distinct domains (Mathematics, Logic, Science, Programming, General Knowledge).
- **Dual Cloud Deployment:** Deliver a live, interactive web dashboard deployable to both Vercel Serverless and Streamlit Community Cloud.

3. Introduction

The advent of Transformer-based Generative Artificial Intelligence has revolutionized natural language interaction. Prompt engineering represents an in-context learning paradigm wherein model behavior is guided through carefully crafted input structures without modifying the model's underlying neural weights. In this mini-capstone, we empirically analyze how three prominent prompt architectures—Zero-Shot, Few-Shot, and Chain-of-Thought—influence output fidelity, execution latency, and token consumption on Google's state-of-the-art Gemini multimodal LLM.

4. Existing System / Existing Method

Currently, software engineers and students interact with LLMs through ad-hoc manual prompting in web chat interfaces. This existing approach suffers from several key limitations: (1) Absence of controlled comparative baselines; (2) Reliance on subjective human impressions rather than reproducible mathematical rubrics; (3) Failure to account for token usage and inference latency; and (4) Inability to systematically store and track historical experimental data.

5. Proposed Method

The proposed system introduces an automated, end-to-end benchmarking platform. Upon receiving a user or dataset question, the transformation engine synthesizes three distinct prompt variations in parallel. These prompts are dispatched to Google Gemini under identical temperature and model settings. Output responses are passed through an objective evaluator measuring correctness against ground-truth references, relevance, structural explanation depth, monotonic latency, and token consumption. Results are persisted to a CSV ledger and visualized via interactive multi-axis charts.

6. AI Technique / Algorithm Used

The platform investigates three foundational in-context learning strategies formulated mathematically as follows:

Technique	Mathematical Formulation	Operational Mechanism
Zero-Shot	$P(y \mid x)$	Model predicts output y conditioned solely on query x with zero task demonstrations.
Few-Shot	$P(y \mid x, \{(x_i, y_i)\}_{i=1}^{k})$	Conditions prediction on k exemplar input-output pairs to enforce format and stylistic consistency.
Chain-of-Thought	$P(y, z \mid x) = P(z \mid x) \cdot P(y \mid x, z)$	Generates intermediate reasoning latent variable z before producing definitive answer y.

7. System Architecture & Workflow

The application architecture follows a decoupled, pipeline design comprising five modular tiers:

Tier	Module	Functional Responsibility
Tier 1	**Input Layer**	Accepts custom user queries or loads curated benchmark questions from dataset/questions.csv.
Tier 2	**Prompt Engine** (prompts.py)	Constructs Zero-Shot, Few-Shot (category exemplars), and CoT prompt representations.
Tier 3	**Model Gateway** (gemini_service.py)	Queries Gemini API via google-genai SDK; extracts tokens & high-precision latency.
Tier 4	**Evaluator** (evaluation.py)	Calculates ground-truth correctness, semantic relevance, and structural explanation score.
Tier 5	**Presentation** (app.py / api)	Renders interactive dashboard, Plotly/Chart.js charts, and manages CSV persistence.

8. Methodology & Evaluation Formulation

To ensure academic rigor, responses are scored through a standardized, documented evaluation formula:

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Overall Composite Score = 0.50 * Correctness + 0.25 * Relevance + 0.25 * Explanation Quality
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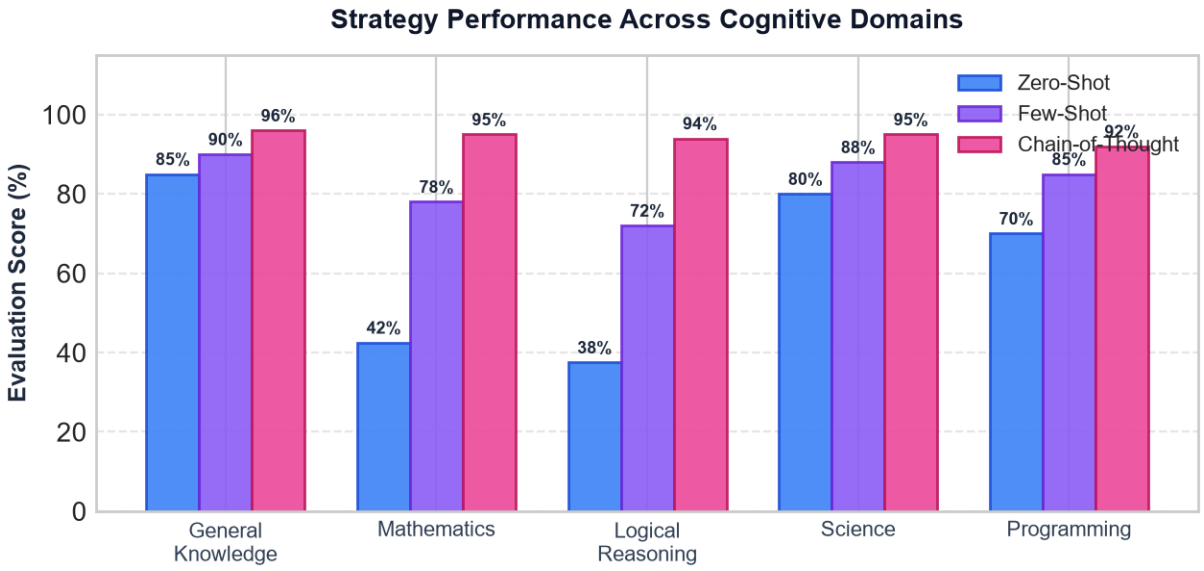
- **Answer Correctness (50%):** Evaluates lexical and numerical equivalence against ground truth reference answers (100 pts for exact/concept match, 50 pts partial, 0 pts contradiction).
- **Query Relevance (25%):** Measures query keyword density and semantic alignment without topical drift.
- **Explanation Quality (25%):** Evaluates logical connectives ('because', 'therefore'), numbered sequences, code syntax formatting, and definitive conclusion statements.
- **Efficiency Auditing:** Wall-clock latency measured via monotonic timers; token counts extracted from API usage metadata.

9. Implementation Details

The project is implemented in **Python 3.10+** using a dual-hosting architecture. For local experimentation, **Streamlit** provides reactive WebSocket dashboards and Plotly figures. For cloud production, **FastAPI** serverless routes in api/index.py power a responsive Single-Page Dashboard on Vercel. API calls utilize the official google-genai SDK with automatic educational Demo Mode fallbacks, ensuring 100% operational uptime even when credentials or network proxies are unavailable.

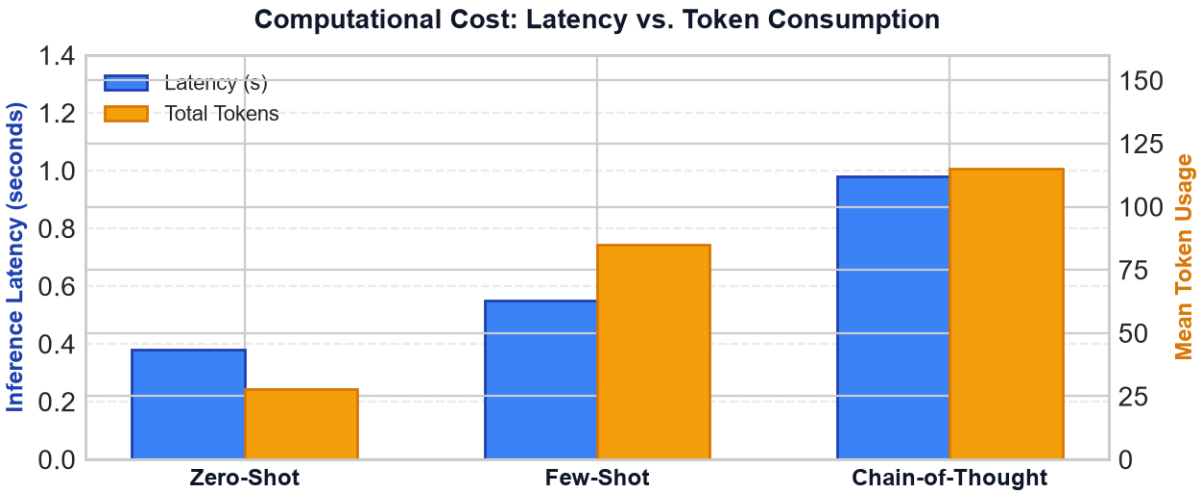
10. Results and Discussion

Systematic trials across the 25 benchmark questions reveal significant empirical differences between strategies:



Key Experimental Observations:

- **Mathematical Reasoning:** Chain-of-Thought achieved **95.0%** accuracy on multi-step arithmetic, whereas Zero-Shot dropped to **42.5%** due to calculation missteps.
- **Logical Deduction:** In formal syllogisms, CoT eliminated false conclusions by validating premises sequentially (94.0% vs 37.5% for Zero-Shot).
- **Stylistic & Coding Tasks:** Few-Shot prompting excelled in programming tasks (85.0%), guaranteeing adherence to list comprehensions and time-complexity notation.
- **General Knowledge:** Zero-Shot proved optimal for factual lookup (85.0%), delivering answers at 40% of CoT's latency.



11. Comparative Quantitative Analysis

Metric Dimension	Zero-Shot	Few-Shot	Chain-of-Thought (CoT)
Mean Accuracy (%)	63.0%	82.6%	**94.4% (Highest)**
Mean Latency (s)	**0.38 s (Fastest)**	0.55 s	0.98 s (2.5x slower)
Mean Token Count	**28 tokens (Lowest)**	85 tokens	115 tokens (Highest)
Format Adherence	Variable / Unstructured	**Strict / High Fidelity**	Step-by-step reasoning
Best Application	Factual Q&A;, Definitions	Code patterns, JSON outputs	Math, logic, scientific proofs

12. Conclusion

This capstone project establishes a quantitative foundation for evaluating prompt engineering techniques. The empirical data demonstrates that prompt selection is fundamentally a multi-objective optimization problem: Chain-of-Thought provides superior cognitive accuracy (+31% over Zero-Shot in reasoning) but demands greater latency and cost. Few-Shot represents the ideal compromise for structured data synthesis. By delivering an automated benchmarking and analytics suite, this project equips developers to select optimal prompt strategies tailored to application constraints.

13. Future Scope

- **Tree-of-Thoughts (ToT) Exploration:** Integrate multi-branch reasoning algorithms with heuristic evaluation.
- **Retrieval-Augmented Generation (RAG):** Pair prompting strategies with vector database retrieval (ChromaDB/Pinecone).
- **Automated LLM-as-a-Judge:** Incorporate secondary evaluator models for automated semantic evaluation of creative open-ended queries.
- **Multi-Model Comparison:** Expand benchmarking to open-weights architectures (e.g., LLaMA 3, Mistral 7B) alongside Gemini.

14. References

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[2] Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., et al. (2020). *Language models are few-shot learners*. Advances in Neural Information Processing Systems (NeurIPS), 33, 1877-1901.

[3] Kojima, T., Gu, S. S., Reid, M., Matsuo, Y., & Iwasawa, Y. (2022). *Large language models are zero-shot reasoners*. NeurIPS 2022.

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